PROJECT-BASED LEARNING

PHASE-I EVALUATION

Secure Marksheet Verifier

Team ID:DSCPP-III-2026-T155

Department of Computer Science & Engineering
Graphic Era (Deemed to be University), Dehradun
Academic Session 2026–27

Team & Mentor Details

Team Information

Team ID: DSCPP-III-2026-T155
Semester: 3rd
Section: A,ML1, ML2, ML4
Domain: *Secure Marksheet Verifier*

Team Members

1. *Diksha Rawat – 2510371956*
2. *Aakhya Gupta – 2510380033*
3. *Divya Singh – 2510370275*
4. *Jeevika Srivastava – 2510370298*

Mentor

Mr. Kamal Kumar Gola

Interactions completed: 3

Problem Statement & Motivation

Problem Statement

Third parties have no quick, secure way to verify if a physical marksheet is tampered with, without accessing the university's full private database.

Motivation

- *Physical marksheets are easily forged or altered.*
- *Full database access risks exposing other records.*

Target Users / Stakeholders

- *1. Recruiters and HR managers verifying candi*
- *Scholarship evaluators confirming applicants' a*
- *3. University administrators publishing certifie hashes.*

Objectives

Project Objectives

1. *Build a Merkle Hash Tree for tamper-proof record storage.*
2. *Design a custom hash engine.*
3. *Enable OCR-based extraction from photographed physical marksheets.*
4. *Implement O(log N) Merkle Proofs for fast verification.*
5. *Perform field-by-field audits to flag tampered subjects.*
6. *Maintain CSV logs for records, ledger, and audits.*

Key Objective: *To instantly verify physical marksheets for tampering using a Merkle Hash Tree and Photo OCR, without exposing the university's full database.*

Proposed Solution

Proposed Approach

1. *University builds Merkle Tree from certified student records.*
2. *Publishes a single immutable Merkle Root Hash.*
3. *Third party photographs the physical marksheet for verification.*
4. *OCR engine extracts roll number and subjec marks.*
5. *System validates data via Merkle Proof and field audit.*

Key Features

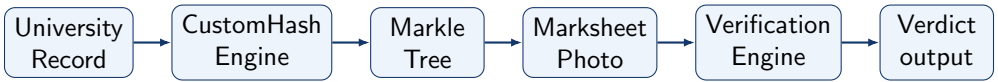
- *Custom dual hash engine*
- *Tamper-proof Merkle Tree with O(log N) proofs*
- *Photo OCR for physical marksheet scanning*
- *Field-by-field tamper detection and reporting*

Integration of PBL Course Concepts

Course	Concepts Applied	Project Module
Data Structures in C	*Merkle Tree, Hash Table , Vectors, Recursion, Custom Hashing*	*Secure record storage, hash generation, Merkle Tree construction, verification proof*
OOPs with C++	*Classes, Objects, Encapsulation, Abstraction, Structs, File I/O*	*Marksheets OCR processing, tamper detection, verification and audit logging*

Requirement: the project combines data structures, algorithms and C++ implementation.

System Architecture / Workflow



Major Components

- *CustomHash Engine*
- *Merkle Tree Module*
- *OCR Engine*
- *Verification Engine*

Data / Control Flow

- *University hashes records into Merkle Tree.*
- *OCR extracts marks from marksheet photo.*
- *Verifier checks proof and flags tampering.*

Technology Stack & Methodology

Technology Stack

- Programming: *C++*
- Library: *C++ Standard Library*
- Data structures: *Merkle Tree, Hash Table (unordered map), Vectors, Recursion, Custom Hashing*
- Algorithm: *FNV-1a and DJB2 Hashing Algorithm,Bottom-up Merkle Tree Construction*
- Version Control: *GitHub*

Development Methodology

1. Design phase — planning modules and data flow.
2. Implementation — coding hash, tree, OCR, verifier.
3. C++ implementationUnit testing — validating each module independently.
4. Viva demo — presenting working system end-to-end.

Initial Progress & Team Contribution

Progress Achieved

- *Project proposal and motivation defined.*
- *Goals and milestones identified.*
- *Core data structures and algorithm selected.*

Individual Contribution

Member	Role	Contribution
Diksha Rawat	*Lead/Developer*	*Team leadership, project coordination and techical contribution*
Aakhya Gupta	*Developer*	*Proof path handling and Merkle proof generation*
Divya Singh	*Developer/Docs*	*Implemented verification logic, report and PPT*
Jeevika Srivastava	*Developer*	*Verification engine, field-by-field tamper audit, and testing*

Roadmap, Expected Outcomes & References

Roadmap

1. Phase-I: Proposal & Design
2. Phase-II: Development
3. Phase-III: Final Implementation

Expected Outcomes

- *Instant verification of physical marksheets achieved.*
- *reliable tamper detection via hash avalanche effect.*
- *Privacy preserved without full database access.*

References

1. *C++ reference.*
2. *Merkle, R. C. (1987). A Digital Signature Based on a Conventional Encryption Function. CRYPTO '87.*
3. *Fowler, G., Noll, L. C., Vo, K. P. — FNV Hash Algorithm.*
4. *Bernstein, D. J. — DJB2 Hash Function, comp.lang.c.*
5. *Course notes*

Thank You